

2

Probability

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2.4 Conditional Probability

Conditional Probability

The probabilities assigned to various events depend on what is known about the experimental situation when the assignment is made.

Subsequent to the initial assignment, partial information relevant to the outcome of the experiment may become available. Such information may cause us to revise some of our probability assignments.

For a particular event A , we have used $P(A)$ to represent the probability, assigned to A ; we now think of $P(A)$ as the original, or unconditional probability, of the event A .

Conditional Probability

In this section, we examine how the information “an event B has occurred” affects the probability assigned to A .

For example, A might refer to an individual having a particular disease in the presence of certain symptoms.

If a blood test is performed on the individual and the result is negative ($B = 5$ negative blood test), then the probability of having the disease will change (it should decrease, but not usually to zero, since blood tests are not infallible).

Conditional Probability

We will use the notation $P(A | B)$ to represent the **conditional probability of A given that the event B has occurred**. B is the “conditioning event.”

As an example, consider the event A that a randomly selected student at your university obtained all desired classes during the previous term’s registration cycle. Presumably $P(A)$ is not very large.

However, suppose the selected student is an athlete who gets special registration priority (the event B). Then $P(A | B)$ should be substantially larger than $P(A)$, although perhaps still not close to 1.

Example 24

Complex components are assembled in a plant that uses two different assembly lines, A and A' .

Line A uses older equipment than A' , so it is somewhat slower and less reliable.

Suppose on a given day line A has assembled 8 components, of which 2 have been identified as defective (B) and 6 as nondefective (B'), whereas A' has produced 1 defective and 9 nondefective components.

Example 24

cont'd

This information is summarized in the accompanying table.

		Condition	
		<i>B</i>	<i>B'</i>
Line	<i>A</i>	2	6
	<i>A'</i>	1	9

Unaware of this information, the sales manager randomly selects 1 of these 18 components for a demonstration.

Prior to the demonstration

$$P(\text{line } A \text{ component selected}) = P(A) = \frac{N(A)}{N} = .44$$

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Example 24

cont'd

However, if the chosen component turns out to be defective, then the event B has occurred, so the component must have been 1 of the 3 in the B column of the table.

Since these 3 components are equally likely among themselves after B has occurred,

$$P(A|B) = \frac{2}{3} = \frac{2/18}{3/18} = \frac{P(A \cap B)}{P(B)} \quad (2.2)$$

Conditional Probability

In Equation (2.2), the conditional probability is expressed as a ratio of unconditional probabilities: The numerator is the probability of the intersection of the two events, whereas the denominator is the probability of the conditioning event B . A Venn diagram illuminates this relationship (Figure 2.8).

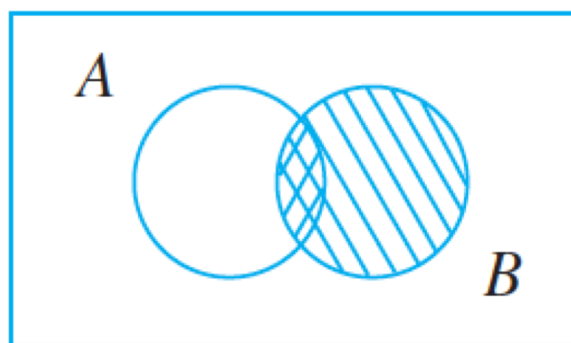


Figure 2.8 Motivating the definition of conditional probability

Conditional Probability

Given that B has occurred, the relevant sample space is no longer S but consists of outcomes in B ; A has occurred if and only if one of the outcomes in the intersection occurred, so the conditional probability of A given B is proportional to $P(A \cap B)$.

The proportionality constant $1/P(B)$ is used to ensure that the probability $P(B|B)$ of the new sample space B equals 1.

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The Definition of Conditional Probability

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The Definition of Conditional Probability

Example 2.24 demonstrates that when outcomes are equally likely, computation of conditional probabilities can be based on intuition.

When experiments are more complicated, though, intuition may fail us, so a general definition of conditional probability is needed that will yield intuitive answers in simple problems.

The Venn diagram and Equation (2.2) suggest how to proceed.

The Definition of Conditional Probability

DEFINITION

For any two events A and B with $P(B) > 0$, the **conditional probability of A given that B has occurred** is defined by

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \quad (2.3)$$

Example 25

Suppose that of all individuals buying a certain digital camera, 60% include an optional memory card in their purchase, 40% include an extra battery, and 30% include both a card and battery. Consider randomly selecting a buyer and let

$A = \{\text{memory card purchased}\}$ and
 $B = \{\text{battery purchased}\}.$

Then $P(A) = .60$,
 $P(B) = .40$, $P(\text{both purchased}) = P(A \cap B) = .30$

Example 25

cont'd

Given that the selected individual purchased an extra battery, the probability that an optional card was also purchased is

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{.30}{.40} = .75$$

That is, of all those purchasing an extra battery, 75% purchased an optional memory card. Similarly,

$$P(\text{battery} | \text{memory card}) = P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{.30}{.60} = .50$$

Notice that in this example:

$$P(A|B) \neq P(A) \text{ and } P(B|A) \neq P(B).$$



The Multiplication Rule for $P(A \cap B)$

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The Multiplication Rule for $P(A \cap B)$

The definition of conditional probability yields the following result, obtained by multiplying both sides of Equation (2.3) by $P(B)$.

The Multiplication Rule

$$P(A \cap B) = P(A|B) \cdot P(B)$$

This rule is important because it is often the case that $P(A \cap B)$ is desired, whereas both $P(B)$ and $P(A|B)$ can be specified from the problem description.

Consideration of $P(B|A)$ gives $P(A \cap B) = P(B|A) \cdot P(A)$

Example 27

Four individuals have responded to a request by a blood bank for blood donations. None of them has donated before, so their blood types are unknown. Suppose only type O+ is desired and only one of the four actually has this type. If the potential donors are selected in random order for typing, what is the probability that at least three individuals must be typed to obtain the desired type?

Making the identification

$B = \{\text{first type not O+}\}$ and

$A = \{\text{second type not O+}\}, P(B) = \frac{3}{4}.$

Example 27

cont'd

Given that the first type is not O+, two of the three individuals left are not O+, so $P(A|B) = \frac{2}{3}$.

The multiplication rule now gives

$$P(\text{at least three individuals are typed}) = P(A \cap B)$$

$$= P(A|B) \cdot P(B)$$

$$= \frac{2}{3} \cdot \frac{3}{4} = \frac{6}{12}$$

$$= .5$$

Example 27

cont'd

The multiplication rule is most useful when the experiment consists of several stages in succession.

The conditioning event B then describes the outcome of the first stage and A the outcome of the second, so that $P(A|B)$ —conditioning on what occurs first—will often be known.

The multiplication rule is easily extended to experiments involving more than two stages.

Example 27

cont'd

For example,

$$\begin{aligned} P(A_1 \cap A_2 \cap A_3) &= P(A_3 | A_1 \cap A_2) \cdot P(A_1 \cap A_2) \\ &= P(A_3 | A_1 \cap A_2) \cdot P(A_2 | A_1) \cdot P(A_1) \quad (2.4) \end{aligned}$$

where A_1 occurs first, followed by A_2 , and finally A_3 .

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Example 26

A news magazine publishes three columns entitled “Art” (A), “Books” (B), and “Cinema” (C). Reading habits of a randomly selected reader with respect to these columns are

<i>Read regularly</i>	A	B	C	$A \cap B$	$A \cap C$	$B \cap C$	$A \cap B \cap C$
<i>Probability</i>	.14	.23	.37	.08	.09	.13	.05

Calculate the following probabilities.

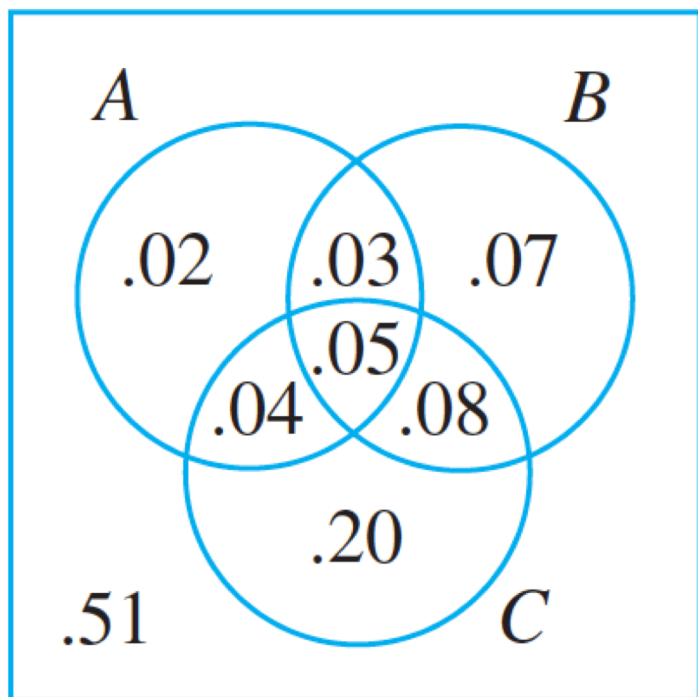
$$P(A|B)$$

$$P(A|B \cup C)$$

$$P(A \cup B|C)$$

Example 26

cont'd



$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{.08}{.23} = .348$$

$$P(A|B \cup C) = \frac{P(A \cap (B \cup C))}{P(B \cup C)} = \frac{.04 + .05 + .03}{.47} = \frac{.12}{.47} = .255$$

$$P(A \cup B|C) = \frac{P((A \cup B) \cap C)}{P(C)} = \frac{.04 + .05 + .08}{.37} = .459$$



Bayes' Theorem

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Bayes' Theorem

The computation of **a posterior probability** $P(A_j|B)$ from given **prior probabilities** $P(A_i)$ and conditional probabilities $P(B|A_i)$ occupies a central position in elementary probability.

The general rule for such computations, which is really just a simple application of the multiplication rule, goes back to Reverend Thomas Bayes, who lived in the eighteenth century.

To state it we first need another result. Recall that events $A_1, \dots, A_k$ are **mutually exclusive** if no two have any common outcomes. The events are **exhaustive** if one A_i must occur, so that $A_1 \cup \dots \cup A_k = \mathcal{S}$.

Bayes' Theorem

The Law of Total Probability

Let $A_1, \dots, A_k$ be mutually exclusive and exhaustive events. Then for any other event B ,

$$P(B) = P(B|A_1)P(A_1) + \dots + P(B|A_k)P(A_k)$$

$$= \sum_{i=1}^k P(B|A_i)P(A_i) \quad (2.5)$$

The Law of Total Probability

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Let $A_1, \dots, A_k$ be mutually exclusive and exhaustive events. Then for any other event B ,

$$\begin{aligned} P(B) &= P(B|A_1)P(A_1) + \dots + P(B|A_k)P(A_k) \\ &= \sum_{i=1}^k P(B|A_i)P(A_i) \end{aligned} \tag{2.5}$$

The Law of Total Probability

$$P(B) = \sum_{i=1}^k P(A_i \cap B) = \sum_{i=1}^k P(B|A_i)P(A_i)$$

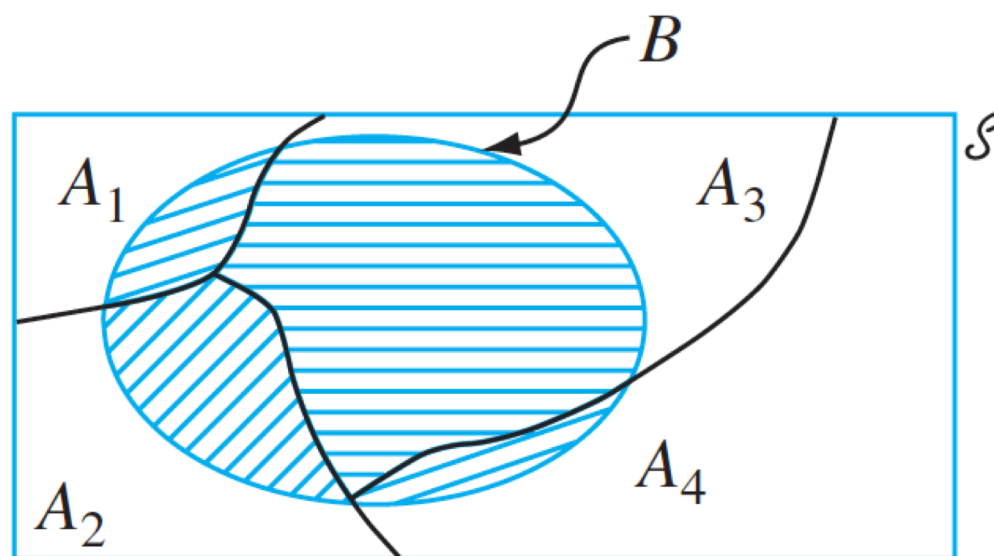


Figure 2.11 Partition of B by mutually exclusive and exhaustive A_i 's

Example 30

An individual has 3 different email accounts. Most of her messages, in fact 70%, come into account #1, whereas 20% come into account #2 and the remaining 10% into account #3.

Of the messages into account #1, only 1% are spam, whereas the corresponding percentages for accounts #2 and #3 are 2% and 5%, respectively.

What is the probability that a randomly selected message is spam?

Example 30

cont'd

To answer this question, let's first establish some notation:

$A_i = \{\text{message is from account \# } i\}$ for $i = 1, 2, 3$,

$B = \{\text{message is spam}\}$

Then the given percentages imply that

$$P(A_1) = .70, P(A_2) = .20, P(A_3) = .10$$

$$P(B|A_1) = .01, P(B|A_2) = .02, P(B|A_3) = .05$$

Example 30

cont'd

Now it is simply a matter of substituting into the equation for the law of total probability:

$$P(B) = (.01)(.70) + (.02)(.20) + (.05)(.10) = .016$$

In the long run, 1.6% of this individual's messages will be spam.

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Bayes' Theorem

Bayes' Theorem

Let $A_1, A_2, \dots, A_k$ be a collection of *k mutually exclusive and exhaustive events with prior probabilities* $P(A_i)$ ($i = 1, \dots, k$).

Then for any other event B for which $P(B) > 0$, *the posterior probability of A_j given that B has occurred* is

$$P(A_j|B) = \frac{P(A_j \cap B)}{P(B)} = \frac{P(B|A_j)P(A_j)}{\sum_{i=1}^k P(B|A_i) \cdot P(A_i)} \quad j = 1, \dots, k \quad (2.6)$$

Bayes' Theorem

Bayes' Theorem

Let $A_1, A_2, \dots, A_k$ be a collection of k mutually exclusive and exhaustive events with *prior* probabilities $P(A_i)$ ($i = 1, \dots, k$). Then for any other event B for which $P(B) > 0$, the *posterior* probability of A_j given that B has occurred is

$$P(A_j|B) = \frac{P(A_j \cap B)}{P(B)} = \frac{P(B|A_j)P(A_j)}{\sum_{i=1}^k P(B|A_i) \cdot P(A_i)} \quad j = 1, \dots, k \quad (2.6)$$

Bayes' Theorem

The transition from the second to the third expression in (2.6) rests on using the multiplication rule in the numerator and the law of total probability in the denominator.

The proliferation of events and subscripts in (2.6) can be a bit intimidating to probability newcomers.

As long as there are relatively few events in the partition, a tree diagram (as in Example 29) can be used as a basis for calculating posterior probabilities without ever referring explicitly to Bayes' theorem.

Example 31

Incidence of a rare disease.

Only 1 in 1000 adults is afflicted with a rare disease for which a diagnostic test has been developed. The test is such that when an individual actually has the disease, a positive result will occur 99% of the time, whereas an individual without the disease will show a positive test result only 2% of the time.

If a randomly selected individual is tested and the result is positive, what is the probability that the individual has the disease?

Example 31

cont'd

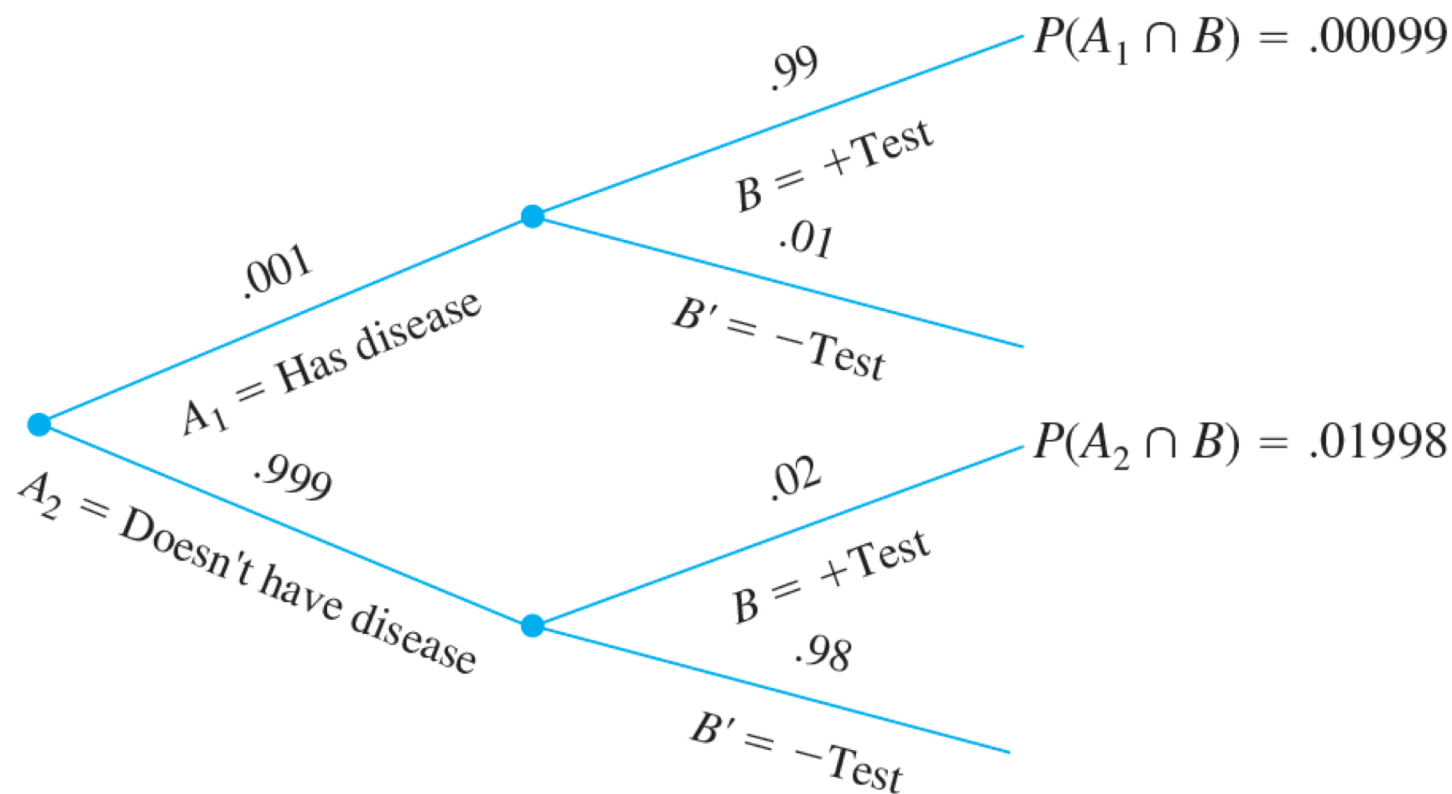


Figure 2.12 Tree diagram for the rare-disease problem

$$P(B) = .00099 + .01998 = .02097,$$

$$P(A_1|B) = \frac{P(A_1 \cap B)}{P(B)} = \frac{.00099}{.02097} = .047$$